Engineering Mathematics - II

1. Sol

$$y = mx + c$$

$$\frac{dy}{dx} = m \quad [\because \text{differential unit constants gone } (m, c)$$

$$\frac{d^2y}{dx^2} = 0$$

Justification:

* The highest derivate is $\frac{d^2y}{dx^2}$
* So, order is 2

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2. Sol

$$y = e^{bx}(a\cos x)$$

* There are two arbitrary constants: a and b
* To eliminate those two, we need to differentiate the equation twice
* order of resulting differential eqn : 2

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3. Sol

Given: A ball is dropped from rest

$\therefore$ rate of velocity is equal to gravity (g)

$$\frac{dv}{dt} = g$$

To find v, integrate with respect to t

$$dv = g\,dt \Rightarrow \int dv = g\int dt$$

$$v = gt + c$$

at $t=0$, $v=0$ (bcz it is at rest initially)

$$0 = g(0) + c \Rightarrow \boxed{c = 0}$$

$$\therefore \boxed{v = gt}$$

Result:

differential eqn : $\frac{dv}{dt} = g$

velocity: $v = gt$

4. Sol

Newton's law of cooling

$$\frac{dT}{dt} \propto (T - T_a)$$

Here, $T_a = 25°C$

$$\frac{dT}{dt} \propto (T-25)$$

$$\boxed{\frac{dT}{dt} = -K(T-25)}$$

[-K bcz it is cooling

5. Sol

$$y = \frac{a}{x} + b$$

a, b → two arbitary constants [∴ aims to remove it

$$\frac{dy}{dx} = \frac{-a}{x^2} \Rightarrow \boxed{a = -x^2 \frac{dy}{dx}} \quad \text{—} (1)$$

$$\boxed{\frac{d^2y}{dx^2} = \frac{+2a}{x^3}} \quad \text{—} (2)$$

sub eqn (1) in eqn (2)

$$\frac{d^2y}{dx^2} = \frac{2\left(-x^2\frac{dy}{dx}\right)}{x^3} \Rightarrow \frac{d^2y}{dx^2} = \frac{-2}{x}\frac{dy}{dx}$$

Result: $\frac{d^2y}{dx^2} + \frac{2}{x}\frac{dy}{dx} = 0$

Interpertation:

velocity varies with position x and decreases as distance increases

6. Sol

* order : 2
* Justification
  - There are two arbitrary constants a and b.
  - To eliminate them, we differentiate twice which resulting differential eqn is of order 2

7. Sol According to Newton's law of cooling,

~~Given:~~ $T - T_s = (T_0 - T_s)e^{-kt}$ — ①

Given: $T_s = 20°C$, $T_0 = 90°$, $T = 70°C$, $t = 8$ min

subtitute above values in eqn ①

$$70 - 20 = (90 - 20)e^{-k(8)}$$

$$50 = 70 e^{-8k} \Rightarrow e^{-8k} = \frac{50}{70} = \frac{5}{7}$$

$$e^{-8k} = \frac{5}{7} \Rightarrow -8k = \ln\left(\frac{5}{7}\right) \Rightarrow 8k = -\ln\left(\frac{5}{7}\right)$$

$$8k = \ln\left(\frac{7}{5}\right) \Rightarrow \boxed{k = \frac{1}{8}\ln\left(\frac{7}{5}\right)} \text{ per minute}$$

8. Sol

free fall means acceleration due to gravity

$$\therefore a(t) = g$$

$$\frac{dv}{dt} = g$$

$$dv = g\,dt$$

integrate to get v

$$\int dv = g\int dt \Rightarrow v = gt + c$$

Given at $t = 0, v = 0$

$$\therefore \boxed{c = 0}$$

Result: * $v = gt$

* velocity increases linearly with time under free fall

9. sol

$$\frac{dT}{dt} = -k(T-T_a)$$

$$\frac{dT}{T-T_a} = -k\,dt$$

integrate on both side

$$\int \frac{dT}{T-T_a} = -k\int dt \Rightarrow \ln|T-T_a| = -kt + c$$

$$T-T_a = e^{-kt+c} \Rightarrow T-T_a = e^{-kt}e^{c} \Rightarrow T-T_a = Ce^{-kt}$$

$$T-T_a = Ce^{-kt} \Rightarrow \boxed{T = T_a + Ce^{-kt}} \quad \text{—} ①$$

at $t=0$, $T = T_0$

$$\therefore T_0 = T_a + Ce^{-k(0)} \Rightarrow \cancel{T_0 = T_a}$$

$$\therefore T_0 = T_a + C(1) \Rightarrow \boxed{C = T_0 - T_a} \quad \text{—} ②$$

sub eqn ② in ①

$$\boxed{T = T_a + (T_0 - T_a)e^{-kt}}$$

Time to reach safe Temperature T_s [let $T = T_s$

$$T_s = T_a + (T_0 - T_a)e^{-kt}$$

$$\frac{T_s - T_a}{T_0 - T_a} = e^{-kt} \Rightarrow \ln\left(\frac{T_s - T_a}{T_0 - T_a}\right) = -kt$$

$$\ln\left(\frac{T_0 - T_a}{T_s - T_a}\right) = kt \Rightarrow \boxed{t = \frac{1}{k}\ln\left(\frac{T_0 - T_a}{T_s - T_a}\right)}$$

10. Sol

$\frac{dv}{dt} \propto v \Rightarrow \frac{dv}{dt} = -kv$

$\frac{dv}{v} = -k\,dt$

integrate on both side

$\int \frac{dv}{v} = -k\int dt \Rightarrow \ln v = -kt + c$

$\boxed{v = Ce^{-kt}}$

given : at $t=0$, $v = 20$

$20 = Ce^{-k(0)} \Rightarrow \boxed{C = 20}$

$\therefore \boxed{v = 20e^{-kt}}$

at $t = 5$, $v = 10$ m/s

$10 = 20e^{-5k} \Rightarrow e^{-5k} = \frac{10}{20} \Rightarrow e^{-5k} = \frac{1}{2}$

$-5k = \ln\frac{1}{2} \Rightarrow 5k = \ln 2$

$\boxed{k = \frac{\ln 2}{5}}$

$\therefore v = 20e^{-(\frac{\ln 2}{5})t}$

at $t = 10$, $v = ?$

$v = 20e^{-(\frac{\ln 2}{5})(10)} \Rightarrow v = 20e^{-2\ln 2}$

$v = 20e^{\ln 2^{-2}} \Rightarrow v = 20(2^{-2}) = 20\left(\frac{1}{2^2}\right) = 20\left(\frac{1}{4}\right) = 5$

$\boxed{v = 5 \text{ m/s}}$

11. Sol

$$\frac{dh}{dt} = -k\sqrt{h} \Rightarrow \frac{dh}{\sqrt{h}} = -k\,dt$$

$$\frac{dh}{h^{+\frac{1}{2}}} = -k\,dt \Rightarrow h^{-1/2}\,dh = -k\,dt$$

integrate on both side

$$\int h^{-\frac{1}{2}}\,dh = -k\int dt \Rightarrow \frac{h^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} = -kt + c$$

$$\frac{h^{\frac{1}{2}}}{\frac{1}{2}} = -kt + c \Rightarrow 2\sqrt{h} = -kt + c$$

$$\sqrt{h} = \frac{c - kt}{2}$$

$$\boxed{h = \left(\frac{c - kt}{2}\right)^2}$$

at $t = 0$, $h = h_0$

$$\cancel{h = \left(\frac{c - k(0)}{2}\right)^2 \Rightarrow h = h}$$

$$h_0 = \left(\frac{c - k(0)}{2}\right)^2 \Rightarrow h_0 = \left(\frac{c}{2}\right)^2 \Rightarrow h_0 = \frac{c^2}{4}$$

$$c^2 = 4h_0 \Rightarrow \boxed{c = 2\sqrt{h_0}}$$

$$\therefore\ h = \left(\frac{2\sqrt{h_0} - kt}{2}\right)^2 \Rightarrow h = \left(\sqrt{h_0} - \frac{kt}{2}\right)^2$$

initial signal strength, $h = 0$

$$0 = \left(\sqrt{h_0} - \frac{kt}{2}\right)^2 \Rightarrow \sqrt{h_0} = \frac{kt}{2} \Rightarrow \boxed{t = \frac{2\sqrt{h_0}}{k}}$$

* Higher initial signal gives longer range, but growth is slow, not linear

12. Sol

$$\frac{dP}{dt} \propto P \Rightarrow \frac{dP}{dt} = kP$$

$$\frac{dP}{P} = k\,dt$$

integrate on both side

$$\int \frac{dP}{P} = k\int dt \Rightarrow \ln P = kt + c$$

$$P = e^{kt+c} \Rightarrow \boxed{P = Ce^{kt}}$$

at $t=0$, $P=500$

$$500 = Ce^{k(0)} \Rightarrow C = 500$$

$$\therefore \boxed{P = 500\,e^{kt}}$$

at $t=4$, $P=1500$

$$1500 = 500\,e^{4k} \Rightarrow 3 = e^{4k} \Rightarrow \ln 3 = 4k$$

$$k = \frac{\ln 3}{4}$$

$$\therefore P = 500\,e^{\left(\frac{\ln 3}{4}\right)t}$$

at $t=6$,

$$P = 500\,e^{\left(\frac{\ln 3}{4}\right)6} = 500\,e^{\frac{3\ln 3}{2}} = 500\,e^{\ln 3^{\frac{3}{2}}}$$

$$P = 500(3)^{\frac{3}{2}} = 500 \times 5.196$$

$$\boxed{P = 2598}$$

13. Sol

$$\frac{dy}{dx} - \frac{2y}{x+1} = (x+1)^3$$

$$P = \frac{-2}{x+1} \qquad Q = (x+1)^3$$

$$I.F = e^{\int \frac{-2}{x+1}dx} = e^{-2\ln|x+1|} = e^{\ln(x+1)^{-2}}$$

Integrating method:

If $\frac{dy}{dx} + Py = Q$

then $y \times I.F = \int (I.F \times Q)\,dx + C$

where $I.F = e^{\int P\,dx}$

$I.E = (x+1)^{-2}$

$\therefore$ The mistake is integrating factor is not e^{-2x}

it is $(x+1)^{-2}$

$$y(x+1)^{-2} = \int (x+1)^{-2}(x+1)^{3}\,dx + C$$

$$y(x+1)^{-2} = \int (x+1)\,dx + C$$

$$y(x+1)^{-2} = \frac{x^2}{2} + x + C \Rightarrow y(x+1)^{-2} = \frac{1}{2}(x^2+2x+4-4)+C$$

$$y(x+1)^{-2} = \frac{1}{2}(x^2+2x) + C \Rightarrow y(x+1)^{-2} = \frac{(x+2)^2}{2} - 2 + C$$

$$y(x+1)^{-2} = \frac{1}{2}\left((x+2)^2-4\right) + C$$

$$y = \frac{(x+2)^4}{2}$$

$I.E = (x+1)^{-2}$

$\therefore$ The mistake is integrating factor is not e^{-2x} it is $(x+1)^{-2}$

$$y(x+1)^{-2} = \int (x+1)^{-2}(x+1)^{3}\,dx + C$$

$$y(x+1)^{-2} = \int (x+1)\,dx + C$$

$$y(x+1)^{-2} = \frac{(x+1)^2}{2} + C$$

$$\boxed{y = \frac{(x+1)^4}{2} + C(x+1)^2}$$ //

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14. Sol

$$\frac{dP}{dt} \propto P \Rightarrow \frac{dP}{dt} = KP$$

$$\frac{dP}{P} = K\,dt$$

integrate on both side

$$\int \frac{dP}{P} = k\int dt \Rightarrow \ln P = kt + c$$

$$P = e^{kt+c} \Rightarrow P = e^{kt} e^{c}$$

$$\boxed{P = Ce^{kt}}$$

at $t=0$; $P = 5$

$$5 = Ce^{k(0)} \Rightarrow c = 5$$

$$\therefore P = 5e^{kt}$$

at $t = 4$, $P = 50$

$$50 = 5e^{k(4)} \Rightarrow 50 = 5e^{4k}$$

$$e^{4k} = 10 \Rightarrow 4k = \ln 10 \Rightarrow k = \frac{\ln 10}{4}$$

$$\therefore \boxed{P = 5e^{\left(\frac{\ln 10}{4}\right)t}}$$

at $t = 10$

$$P = 5e^{\left(\frac{\ln 10}{4}\right)10} \Rightarrow P = 5e^{\frac{\ln 10}{2}(5)} \Rightarrow P = 5e^{\frac{5}{2}\ln 10} = 5e^{\ln 10^{\frac{5}{2}}}$$

$$\cancel{P = 5e^{\ln 1}} \Rightarrow \cancel{\boxed{P = 5}}$$

$$P = 5(10)^{\frac{5}{2}} \Rightarrow \boxed{P = 1581}$$

Result:

After 10 days, $P = 1581$ students

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15. sol

$$\frac{dy}{dx} + 3y = 2x - 1$$

if $\frac{dy}{dx} + Py = Q$

then $y \times I.F = \int (I.F \times Q)\,dx + C$

where $I.F = e^{\int P dx}$

$P = 3$ $\quad Q = 2x - 1$

$$I.F = e^{\int 3dx} \Rightarrow I.F = e^{3x}$$

$\therefore \; ye^{3x} = \int e^{3x}(2x-1)\,dx + c \longrightarrow (1)$

$\int u\,dv = uv - \int v\,du$

let

$u = 2x - 1 \qquad du = 2dx$

$dv = e^{3x}dx \qquad v = \frac{1}{3}e^{3x}$

$$\int e^{3x}(2x-1) = (2x-1)\left(\frac{1}{3}e^{3x}\right) - \int \frac{1}{3}e^{3x}(2dx)$$

$$\int e^{3x}(2x-1) = (2x-1)\left(\frac{1}{3}e^{3x}\right) - \frac{2}{3}\int e^{3x}dx$$

$$\int e^{3x}(2x-1) = (2x-1)\left(\frac{1}{3}e^{3x}\right) - \frac{2e^{3x}}{9}$$

$$= \frac{2xe^{3x}}{3} - \frac{e^{3x}}{3} - \frac{2e^{3x}}{9} = \frac{2xe^{3x}}{3} - \frac{5}{9}e^{3x}$$

$$\int e^{3x}(2x-1) = \frac{2}{3}xe^{3x} - \frac{5}{9}e^{3x} \longrightarrow (2)$$

Sub eqn (2) in eqn (1)

$$ye^{3x} = \frac{2}{3}xe^{3x} - \frac{5}{9}e^{3x} + c$$

$$\boxed{y = \frac{2}{3}x - \frac{5}{9} + ce^{-3x}}$$

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16. Sol

$\frac{dP}{dt} \propto P \Rightarrow \frac{dP}{dt} = KP$

$\frac{dP}{P} = Kdt$

integrate on both side

$\int \frac{dP}{P} = K\int dt \Rightarrow \ln P = Kt + C \Rightarrow P = e^{Kt+C} = e^{Kt}e^{c}$

$$\boxed{P = Ce^{Kt}}$$

at $t=0$, $P=50$

$$50 = Ce^{k(0)} \Rightarrow C = 50$$

$$\therefore P = 50e^{kt}$$

at $t=2$, $P=200$

$$200 = 50e^{2k} \Rightarrow 4 = e^{2k} \Rightarrow \ln 4 = 2k$$

$$\boxed{k = \frac{\ln 4}{2}}$$

$$\therefore P = 50e^{\left(\frac{\ln 4}{2}\right)t}$$

at $t=6$

$$P = 50e^{\left(\frac{\ln 4}{2}\right)(6)} \Rightarrow P = 50e^{3\ln 4} \Rightarrow P = 50e^{\ln 4^3}$$

$$P = 50(4^3) = 50 \times 64$$

$$P = 3200$$

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17. Sol

$$\frac{d^2y}{dx^2} + 3y = \sin x$$

* It is non-homogenous differential equation
* order = 2
* degree = 1

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18. Sol → The particular integral tells how the system responds directly to external influences in real world

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19. Sol

$$x'' + 25x = 0 \Rightarrow \frac{dx^2}{dy^2} + 25x = 0$$

$$(D^2 + 25)x = 0$$

$$m^2 + 25 = 0 \Rightarrow m^2 = -25 \Rightarrow m = \pm 5i$$

~~∴ frequency~~

$$\therefore \boxed{\omega = 5}$$

$$C.F = C_1 \cos 5t + C_2 \sin 5t$$

$$\boxed{\text{Amplitude} = \sqrt{C_1^2 + C_2^2}}$$

nature → oscillation with exponential decay

frequency

$$\boxed{f = \frac{5}{2\pi}\,\text{Hz}} \Leftarrow f = \frac{\omega}{2\pi}$$

20. Sol

$$\text{frequency}(f) = \frac{\text{number of cycles } (n)}{\text{time } (t)}$$

$f = ?$ $t = 10$, $n = 50$

$f = \frac{50}{10} \Rightarrow \boxed{f = 5\ Hz}$

significiance

Helps prevent resonance and ensure safe, stable system design

21. Sol

$\frac{d^2y}{dx^2} + 4y = 0$ Given: $y(0) = 2$, $y'(0) = -4$

$y = c_1 \cos 2x + c_2 \sin 2x$ —(1) $y' = -2c_1 \sin 2x + 2c_2 \cos 2x$

sub ~~y(o)~~ $y(0) = 2$, $x = 0$

$2 = c_1 \cos 2(0) + c_2 \sin 2(0) \Rightarrow \boxed{2 = c_1}$

sub $y'(0) = -4$

$-4 = -2c_1 \sin 2(0) + 2c_2 \cos 2(0)$

$-4 = 0 + 2c_2 \Rightarrow \boxed{c_2 = -2}$

Result:

* $c_1 = 2$, $c_2 = -2$
* $y = 2\cos 2x - 2\sin 2x$
* behaviour
  * It is homogenous differential eqn
  * The system is ~~underfor~~ underdamped

22. Sol

* Initial conditions fix the exact solution of second order system
* They give the starting values like initial displacement and velocity, which decide how the system begins to respond

23. Sol

$$\frac{d^2q}{dt^2} + \left(\frac{K}{m}\right)q = 0$$

$$\left(D^2 + \frac{K}{m}\right)q = 0$$

$$d^2 + \frac{K}{m} = 0 \Rightarrow d^2 = -\frac{K}{m} \Rightarrow d = \pm \frac{K}{m} i$$

$$C.F = C_1 \cos \frac{K}{m} t + C_2 \sin \frac{K}{m} t$$

$$\therefore \boxed{\omega = \sqrt{\frac{K}{m}}}$$

* If K increases, the value of ω also increases
* ∴ the circuit oscillates faster and the frequency increases

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24. Sol

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

* It represents simple harmonic motion
* because acceleration is proportional to displacement and opposite in direction

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

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25. Sol

$$\frac{d^2i}{dt^2} + 4\frac{di}{dt} + 4i = 0$$

$$(D^2 + 4D + 4)i = 0$$

$$m^2 + 4m + 4 = 0 \Rightarrow (m+2)^2 = 0$$

$$\therefore \text{roots}: m = -2, -2$$

$$C.F = C_1 e^{ax} + C_2 e^{bx} \Rightarrow C.F = C_1 e^{-2x} + C_2 e^{-2x}$$

$$\boxed{C.F = (C_1 + C_2)e^{-2x}}$$

* $\Delta = b^2 - 4ac = 16 - 4(1)(4) \Rightarrow \Delta = 0$

∴ * The system is critically damped
* It faster return to equilibrium

26. Sol

$$\frac{d^2x}{dt^2} + 8\frac{dx}{dt} + 16x = 0$$

$$(D^2 + 8D + 16)x = 0$$

$$m^2 + 8m + 16 = 0 \Rightarrow (m+4)^2 = 0$$

$\therefore$ roots $= -4, -4$

$$C.F = c_1 e^{ax} + c_2 e^{bx} = c_1 e^{-4x} + c_2 e^{-4x}$$

$$\boxed{C.F = (c_1 + c_2)e^{-4x}}$$

$\Delta = b^2 - 4ac = 64 - 4(1)(16) \Rightarrow \Delta = 0$ * The system is real and equal

$\therefore$ * The system is critically damped

* It faster return to equilibrium

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27. Sol

$$\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = 0$$

$$(D^2 + 5D + 6)y = 0$$

Auxiliary eqn: $m^2 + 5m + 6 = 0$

$$(m+2)(m+3) = 0$$

$\therefore$ roots : $-2, -3$

$$C.F = c_1 e^{at} + c_2 e^{bt}$$

$$\boxed{C.F = c_1 e^{-2t} + c_2 e^{-3t}}$$

$\Delta = b^2 - 4ac = 25 - 4(1)(6) \Rightarrow \Delta = 1$ $\Delta > 0$

* The system is real and unequal
* The system is over damped
* Oscillation with exponential decay

28. Sol

$$m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = 0$$

Given: $m = 2$ kg , $c = 6$ Ns/m , $k = 5$ N/m

$$m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = 0$$

$$2\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 5x = 0$$

$$\frac{d^2x}{dt^2} + 3\frac{dx}{dt} + \frac{5}{2}x = 0$$

$$\left(D^2 + 3D + \frac{5}{2}\right)x = 0$$

auxiliary equation: $m^2 + 3m + \frac{5}{2} = 0$

$\Delta = b^2 - 4ac = 3^2 - 4(1)\left(\frac{5}{2}\right) = 9 - 10 \Rightarrow \Delta = -1$ $\quad \Delta < 0$

$\Delta < 0$

$\therefore$ nature: the system is underdamped

* Oscillation with exponential decay

29. Sol

$$\frac{d^2y}{dt^2} + 4y = t^2$$

To find: $y = CF + PI$

$$(D^2+4)y = 0$$

$$m^2 + 4 = 0 \Rightarrow m = \pm 2i$$

$\cancel{CF = c_1 \cos 2t + c_2 \sin 2t}$

$$C.F = c_1 e^{2it} + c_2 e^{-2it}$$

To find: PI

$$PI = \frac{1}{D^2+4} t^2 = \frac{1}{4}\left(\frac{1}{\frac{D^2}{4}+1}\right) t^2$$

$$PI = \frac{1}{4}\left(\frac{D^2}{4}+1\right)^{-1} t^2$$

$$PI = \frac{1}{4}\left(1 - \frac{D^2}{4} + \frac{\left(\frac{D^2}{4}\right)^2}{2} \cdots\right)t^2 \quad \left[\because \text{WKT } (x+1)^{-1} = 1 - x + \frac{x^2}{2} - \frac{x^3}{3} + \cdots\right.$$

$$\cancel{= \frac{1}{4}\left(t^2 - \frac{D^2 t^2}{4}\right)}$$

$$PI = \frac{1}{4}\left(t^2 - \frac{D^2}{4}(t^2) + \frac{D^4}{32}(t^2) - \cdots\right)$$

$$\frac{D^2}{4}(t^2) = \frac{(t^2)''}{4} = \frac{(2t)'}{4} = \frac{2}{4} = \frac{1}{2}$$

$$= \frac{1}{4}\left(t^2 - \frac{2}{4} + 0\right) = \frac{t^2}{4} - \frac{2}{16}$$

$$\boxed{PI = \frac{t^2}{4} - \frac{1}{8}}$$

$\therefore$ The required eqn $y = CF + PI$ is

$$y = c_1 e^{2it} + c_2 e^{-2it} + \frac{t^2}{4} - \frac{1}{8} \quad //$$

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31. Sol

$$F(t) = 8\cos \omega t$$

$$\frac{d^2y}{dx^2} + y = \cos 2x$$

$$(D^2+1)y = 0$$

$$m^2 + 1 = 0$$

$$m = \pm i$$

$$\boxed{C.F = c_1 e^{ix} + c_2 e^{-ix}}$$

$$PI = \frac{1}{D^2+1}\cos 2x \qquad \text{sub } D^2 = -(2)^2$$

$$PI = \frac{1}{-4+1}\cos 2x \Rightarrow \boxed{PI = -\frac{1}{3}\cos 2x}$$

$$y = C.F + PI$$

$$y = c_1 e^{ix} + c_2 e^{-x} - \frac{1}{3}\cos 2x \quad //$$

32. Sol

The forced vibration eqn is

$$m\frac{d^2x}{dt^2}+c\frac{dx}{dt}+kx=F(t)$$

Given: $m=1kg$, $c=4Ns/m$, $k=5N/m$, $F(t)=10\cos 2t$

$$\therefore \boxed{\frac{d^2x}{dt^2}+4\frac{dx}{dt}+5x=10\cos 2t} \quad -(1)$$

$$(D^2+4D+5)=0$$

auxiliary eqn : $m^2+4m+5=0$ $-(2)$

roots : $-2-i, +2+i$ (after solve eqn (2))

$\Delta=b^2-4ac=16-4(1)(5)=-4$

$\Delta<0$

* nature ; the system is underdamped
* oscillation with exponential decay

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33. Sol

$x(t)=t^2$

* The corrected form : $\vec{r}(t)=t^2\hat{i}$
* The error is that position is treated as a scalar quantity
* The position must be represented as a vector because it
* the position must be represented as a vector because it has both magnitude and direction

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34. Sol

$T(x,y)=5x+3y$

* It represents a scalar field because temperature has only magnitude and no direction

35

35. Sol

Given: $\vec{r}(t) = t\hat{i} + t^2\hat{j}$

General form: $\vec{r}(t) = x\hat{i} + y\hat{j}$

$\therefore x = t \quad , y = t^2$

$\therefore y = x^2$

$\frac{dy}{dx} = 2x$

$\boxed{\frac{dy}{dx} = 2t}$

tangent $= \frac{dy}{dx}$

tangent vector ($\vec{v}$)

$\vec{v} = \frac{d\vec{r}}{dt}$

$\vec{v} = \hat{i} + 2t\hat{j}$

* since, the tangent direction varies at different points, the motion is curved, not straight

36. Sol

A particular moving in a circle has constant curvature directed toward the centre

37. Sol

$\vec{v}(t) = 3t\hat{i} + 2t^2\hat{j}$

$\vec{a}(t) = \frac{d\vec{v}}{dt} = \frac{d}{dt}(3t\hat{i} + 2t^2\hat{j})$

$\vec{a}(t) = 3\hat{i} + 4t\hat{j}$

$\boxed{\vec{a}(t) = 3\hat{i} + 4t\hat{j}}$

since acceleration depends on t, it is not constant.

38. Sol

$$\vec{r}(t) = 3t\hat{i} + 2t^2\hat{j}$$

General form: $\vec{r}(t) = x\hat{i} + y\hat{j}$

$\therefore \; x = 3t \; , \; y = 2t^2$

*The motion is curved (parabolic) motion

39. Sol

* The error is assuming tangent and normal vectors are parallel
* They are prependicular to each other.

40. Sol

*The unit tangent vector indicates the direction of motion of a particle at any point on the path.

41. Sol

$$\vec{r}(t) = t^2\hat{i} + 3t\hat{j}$$

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = 2t\hat{i} + 3\hat{j}$$

$$\boxed{\vec{v} = 2t\hat{i} + 3\hat{j}}$$

$$\vec{a}(t) = \frac{d\vec{v}}{dt} = 2\hat{i}$$

$\boxed{\vec{a}(t) = 2\hat{i}}$ acceleration is independent of time (t)

at $t = 2$

$$\vec{v} = 4\hat{i} + 3\hat{j}$$

$$\vec{a} = 2\hat{i}$$

* The motion is uniformly acclerated because accleration remains constant throughout the motion

42. Sol

$$\vec{F} = yz\hat{i} + xz\hat{j} + xy\hat{k}$$

Divergence of a vector field = $\nabla . \vec{F}$

$$\nabla . \vec{F} = \frac{\partial}{\partial x}(yz) + \frac{\partial}{\partial y}(xz) + \frac{\partial}{\partial z}(yk)$$

$$\nabla . \vec{F} = 0 + 0 + 0$$

$$\boxed{\nabla . \vec{F} = 0}$$

* The flow system is incompressible because divergence of the field is zero

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43. Sol

$$\vec{r}(t) = t^3\hat{i} + 2t^2\hat{j} + t\hat{k}$$

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = 3t^2\hat{i} + 4t\hat{j} + \hat{k}$$

$$\vec{a}(t) = \frac{d\vec{v}}{dt} = 6t\hat{i} + 4\hat{j}$$

at $t = 1$

$$\vec{v}(1) = 3\hat{i} + 4\hat{j} + \hat{k}$$

$$\vec{a}(1) = 6\hat{i} + 4\hat{j}$$

Tangential component of accleration:

$$a^T = \frac{\vec{v} . \vec{a}}{|\vec{v}|} = \frac{(3\hat{i} + 4\hat{j} + \hat{k}) . (6\hat{i} + 4\hat{j})}{\sqrt{3^2 + 4^2 + 1^2}}$$

$$= \frac{18 + 16 + 0}{\sqrt{9 + 16 + 1}} = \frac{34}{\sqrt{26}}$$

$$\boxed{a^T = \frac{34}{\sqrt{26}}}$$

44. Sol

$$\vec{F} = 2x\hat{i} + 3y\hat{j}$$

$$\nabla\cdot\vec{F} = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}$$

$$\cancel{\nabla\cdot\vec{F} = 2\hat{i} +}$$

$$\nabla\cdot\vec{F} = 2 + 0 + 0 + 3$$

divergence $\boxed{\nabla\cdot\vec{F} = 5}$

$\nabla\cdot\vec{F} \neq 0$ ∴ The field is not flux conserving

* Physically, the field acts as a source generating outward flux.

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45. Sol

$$V(x, y, z) = xyz$$

Gradient

$$\Delta V = \frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}$$

$$\Delta V = yz\hat{i} + xz\hat{j} + xy\hat{k}$$

at (1, 1, 1)

$$\boxed{\Delta V = \hat{i} + \hat{j} + \hat{k}}$$

$$\vec{a} = \hat{i} + \hat{j} + \hat{k}$$

unit vector in that direction, $\vec{u} = \frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{1^2 + 1^2 + 1^2}}$

$$\vec{u} = \frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$$

directional derivatives $\Big\} = \vec{a}\cdot\vec{u} = (\hat{i} + \hat{j} + \hat{k})\cdot\left(\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})\right) = \frac{1+1+1}{\sqrt{3}} = \frac{3}{\sqrt{3}}$

$$= \sqrt{3}$$

* Physically, electric potential increases at ~~at~~ rate $\sqrt{3}$ in the given direction

46. Sol

$\vec{F} = y\hat{i} - x\hat{j}$

General form: $\vec{F} = P\hat{i} + Q\hat{j}$ $\therefore P = y, Q = -x$

step ②

$\nabla \times \vec{F} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\hat{k}$

step ①

$\left[\text{From } \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & -x & 0 \end{vmatrix}\right.$

$= (-1 - 1)\hat{k}$

$$\boxed{\nabla \times \vec{F} = -2\hat{k}}$$

nature:

$\nabla \times \vec{F} \neq 0$ $\therefore$ The fluid flow is rotational

* Negative sign indicates clockwis rotation about the z-axis

* The rotational motion enhances fluid mixing efficiency

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47. Sol

$T(x, y, z) = x^2 + y^2 + z^2$

$\Delta T = \frac{\partial T}{\partial x}\hat{i} + \frac{\partial T}{\partial y}\hat{j} + \frac{\partial T}{\partial z}\hat{k}$

$\Delta T = 2x\hat{i} + 2y\hat{j} + 2z\hat{k}$

at $(1,1,1)$

$\Delta T = 2\hat{i} + 2\hat{j} + 2\hat{k}$ $\Rightarrow$ $\boxed{\Delta T = 2(\hat{i} + \hat{j} + \hat{k})}$

unite vector in that direction, $\vec{u} = \frac{2\hat{i} + 2\hat{j} + 2\hat{k}}{\sqrt{2^2 + 2^2 + 2^2}} = \frac{2(\hat{i} + \hat{j} + \hat{k})}{\sqrt{4 \times 3}}$

maximum rise direction $\boxed{\vec{u} = \frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})}$

* Physically, the gradient identifies the direction of ~~fast~~ heat increase, helping efficient battery cooling and safety management

48. Sol

$$\vec{F} = 2x\hat{i} + 3y\hat{j} + 4z\hat{k}$$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x & 3y & 4z \end{vmatrix}$$

$$\nabla \times \vec{F} = \left(\frac{\partial}{\partial y}(4z) - \frac{\partial}{\partial z}(3y)\right)\hat{i} - \left(\frac{\partial}{\partial x}(4z) - \frac{\partial}{\partial z}(2x)\right)\hat{j}$$

$$+ \left(\frac{\partial(3y)}{\partial x} - \frac{\partial(2x)}{\partial y}\right)$$

$$\nabla \times \vec{F} = 0\hat{i} - 0\hat{j} + 0k$$

$$\boxed{\nabla \times \vec{F} = \vec{0}}$$

$\therefore$ The feild is conservative because its curl is zero